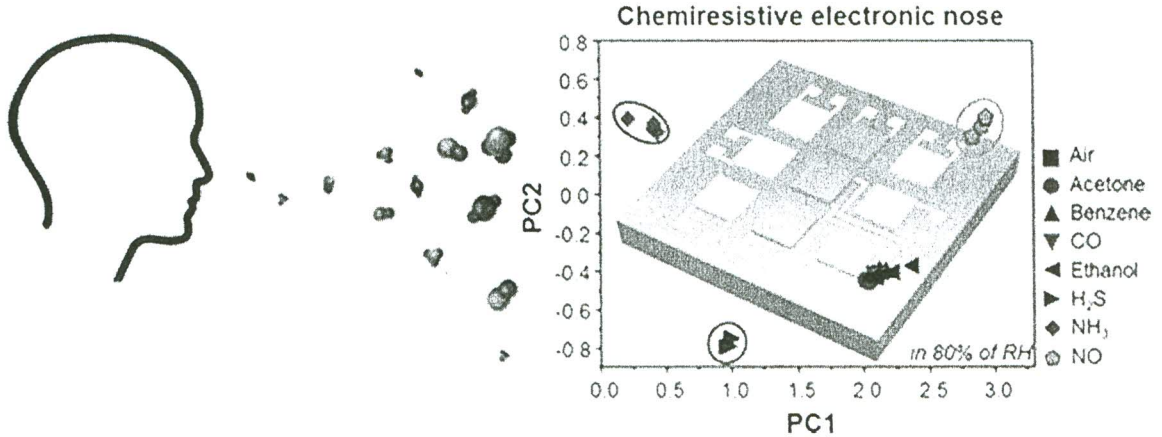
Analytical Chemistry Cumulative Exam
September 30, 2016

This exam is inspired by the History of GCMS article by Hites found in *Analytical Chemistry* News and Features.

1. Structural elucidation of chemical compounds is one of the most important types of chemical analysis that can be performed.
   a. In addition to mass spectrometry, what are the other common methods used for organic compound structural elucidation?
   b. For each of the methods that you have stated in 1a, provide a terse description of the transduction method used for each method and how it "reports" on chemical structural information.
   c. For all methods listed in 1a plus mass spectrometry, explain the most useful chemical information provided or gained by these methods.
   d. Give strengths and limitations of these methods (including mass spectrometry).

2. The article states that by 1956 a time-of-flight (TOF) mass spectrometer was integrated with the GC as a replacement to sector instruments. Fast forward to today and now we see TOF instruments on GC × GCs rather than quadrupoles. One of the primary reasons for this is duty cycle of the mass spectrometer.
   a. Why would duty cycle be an important consideration in GC?
   b. Explain how the duty cycle works for measuring a range of *m/z* values in a TOF vs. a quadrupole mass analyzer.

3. In the field of analytical chemistry, there are always new competing technologies being developed. One technology that has an extensive history is that of the electronic nose. Shown below is an example of a new electronic nose used for breath analysis. The different sites (9 on this chip) respond with different sensitivity and selectivity toward different analytes.
   a. Provide a definition for sensitivity.
   b. Provide a definition for selectivity.
   c. Breath contains more than 4000 components. Electronic noses use multivariate calibration techniques such as Principal Component Regression (PCR) and Partial Least Squares Analysis (PLS).
      i. What is a Principal Component?
      ii. Why is it necessary to use Principal Components or other multivariate calibration techniques for a device such as the electronic nose?
   d. Since breath does contain more than 4000 components. What would be the advantages vs. disadvantages of using GCMS vs. an electronic nose for quantitation of a subset of the 100 most abundant components?